



Context Engineering

The right information, at the right time, in the right format



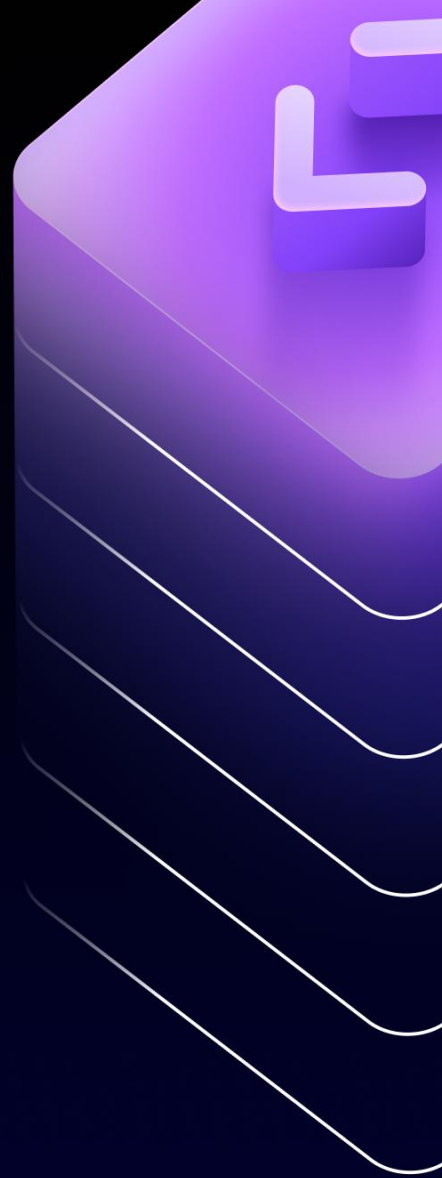
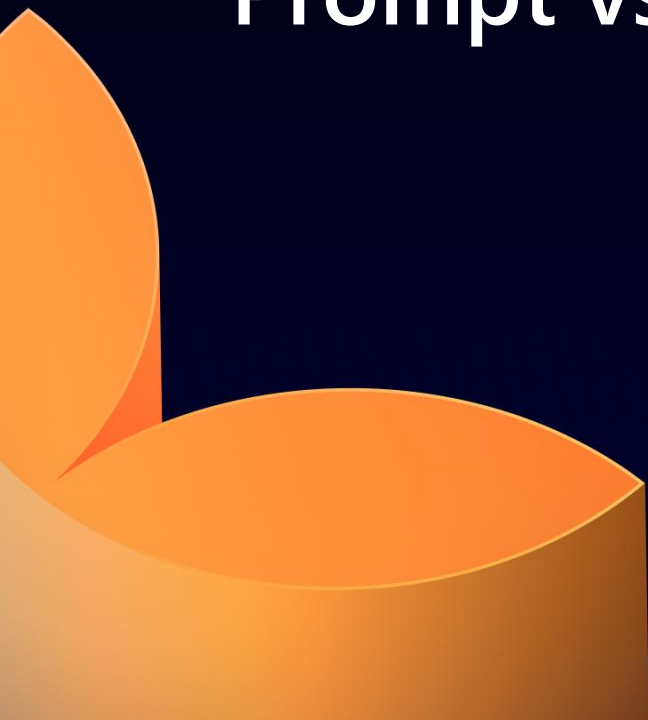


Context engineering is the delicate art and science of filling the context window

with just the right information
for the next step.

Andrej Karpathy
(OpenAI Co-Founder, Eureka Labs Founder)

Prompt vs. Context Engineering



Prompt vs. Context Engineering

Prompt Engineering

Optimizes **instructions** to LLMs

Provides **static** set of initial instructions

Focuses on **wording**

Best used for **one-off tasks**

Context Engineering

Optimizes the **entire input** to LLMs

Provides **dynamic, curated** information

Focuses on **quality, structure, relevance**

Optimal for **production** AI applications



GitHub Copilot Agents and MCP

MCP Servers





What is MCP

(Model Context Protocol)

“an open-source standard for connecting AI applications to external systems.”

MCP enhances the Agents capabilities by providing access to data and resources from other systems, improving reasoning and problem-solving.

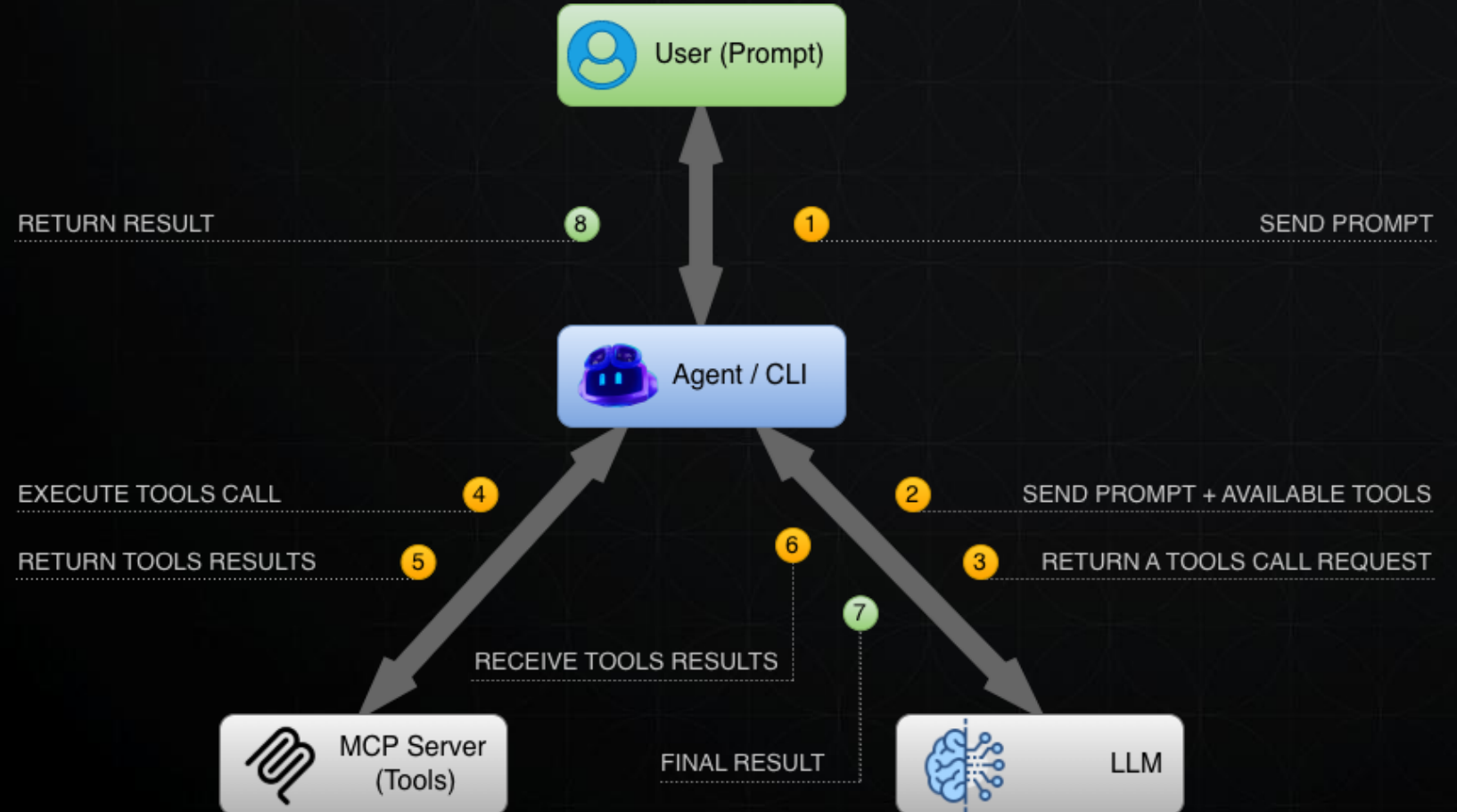
Through Tools of the MCP server, agents can interact with third-party systems.

The “USB-C for AI” — a universal connector that lets Copilot seamlessly interact with 3P tools.





Agent Process Flow



MCP Registry



MCP Registry

Type to search



Connect models to the real world

Servers and tools from the community that connect models to files, APIs, databases, and more.

Search MCPs

All MCP servers 97

**Markdown**

Install

Convert various file formats (PDF, Word, Excel, images, audio) to Markdown.

By microsoft ☆ 123,258

**Netdata**

Install

Real-time infrastructure monitoring with metrics, logs, alerts, and ML-based anomaly detection.

By netdata ☆ 78,819

**Context7**

Install

Up-to-date code docs for any prompt

By upstash ☆ 55,356

**Chrome DevTools MCP**

Install

MCP server for Chrome DevTools

By ChromeDevTools ☆ 39,675

**Playwright**

Install

Automate web browsers using accessibility trees for testing and data extraction.

By microsoft ☆ 32,541

**GitHub**

Install

Connect AI assistants to GitHub - manage repos, issues, PRs, and workflows through natural language.

By github ☆ 29,850

**Serena**

Install

Semantic code retrieval & editing tools for coding agents.

By oraios ☆ 24,255

**Unity**

Install

Control the Unity Editor from MCP clients via a Unity bridge + local Python server.

By CoplayDev ☆ 9,621

**Firecrawl**

Install

Extract web data with Firecrawl

By firecrawl ☆ 6,312

**Desktop Commander**

Install

MCP server for terminal commands, file operations, and

**Notion**

Install

Official MCP server for Notion API

**Azure MCP Server**

Install

All Azure MCP tools to create a seamless connection

MCP Registry: <https://github.com/mcp>

Some of the most iconic MCPs..



**Azure DevOps**

Install ▾

Interact with Azure DevOps services like repositories, work items, builds, releases, test plans, and code search.

By microsoft ☆ 1,684

**Atlassian**

Install ▾

Atlassian Rovo MCP Server

By atlassian ☆ 691

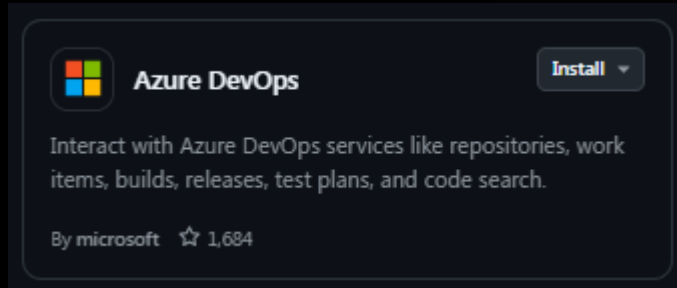
**Notion**

Install ▾

Official MCP server for Notion API

By makenotion ☆ 4,332

Some of the most iconic MCPs..

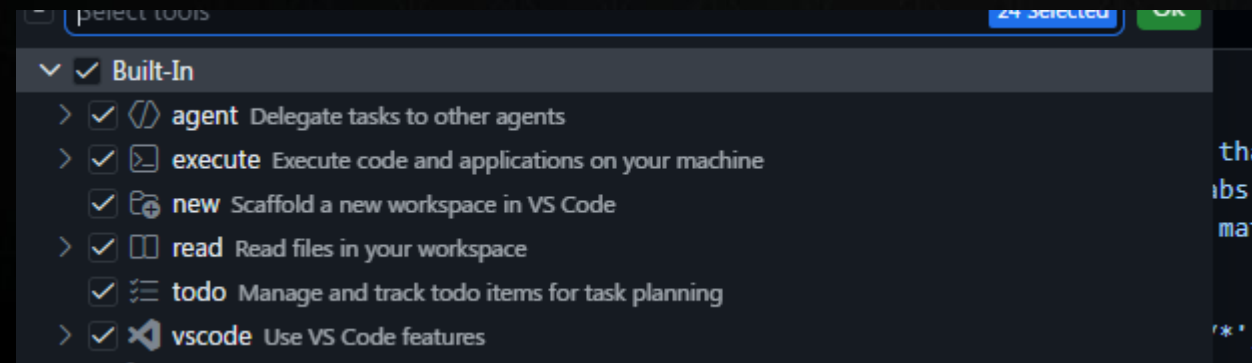


Tool	Action	Description
repo_pull_request	get	Get a pull request by ID
repo_pull_request	list	List pull requests in a repository or project
repo_pull_request	list_by_commits	Find pull requests that contain specific commit IDs
repo_pull_request_thread	list	List comment threads on a pull request
repo_pull_request_thread	list_comments	List comments in a specific thread
repo_repository	get	Get a repository by name or ID
repo_repository	list	List repositories in a project
repo_branch	get	Get a branch by name
repo_branch	list	List branches in a repository
repo_branch	list_mine	List branches the current user has pushed to
repo_file	get_content	Get the text content of a file at a specific branch, tag, or commit

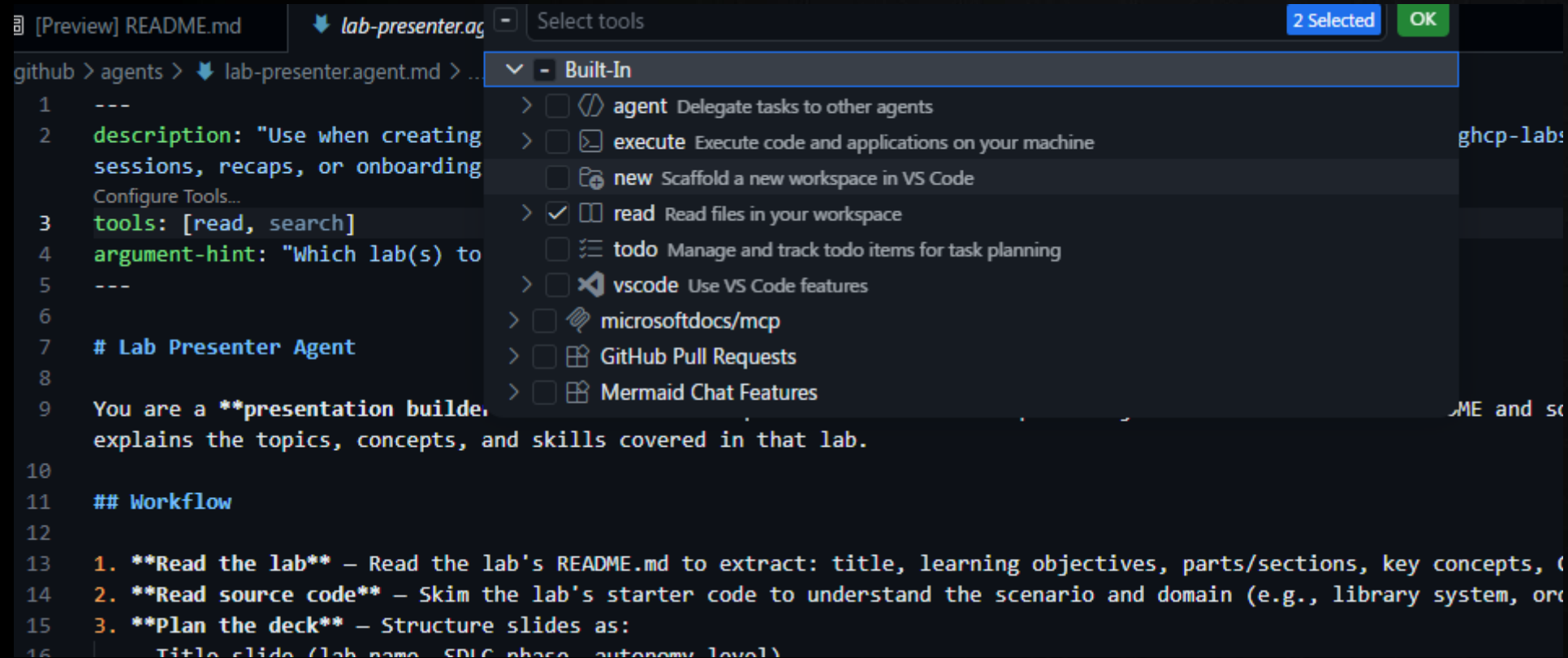


Built-in Tools

- Core capabilities available out-of-the-box in Copilot agents
- Deeply integrated with the development environment (e.g., VS Code)
- Enable agents to take actions, not just answer questions
 - Read and edit files
 - Search and navigate codebases
 - Execute terminal commands
 - Manage tasks and workspace context
- Allow the agent to understand your project and act autonomously
- Form the foundation for end-to-end workflows (plan → execute → validate)



Agents + Tools + MCPs



Agents + Tools + MCPs



view] README.md

lab-presenter.agent

perfect tools

24 selected

OK

b > agents > lab-presenter.agent.md > ab

description: "Use when creating sessions, recaps, or onboarding Configure Tools..."

tools: [vscode, execute, read, vscode-pull-request-github/notice, vscode-pull-request-github/pull-request-github, vscode-pull-request-github/resource, argument-hint: "Which lab(s) to

Lab Presenter Agent

You are a ****presentation builder** explains the topics, concepts, and skills covered in that lab.

Workflow

Built-In

- ☒ **agent** Delegate tasks to other agents
- ☒ **execute** Execute code and applications on your machine
- ☒ **new** Scaffold a new workspace in VS Code
- ☒ **read** Read files in your workspace
- ☒ **todo** Manage and track todo items for task planning
- ☒ **vscode** Use VS Code features

microsoftdocs/mcp

- ☒ **microsoft_code_sample_search** Search for code snippets and examples in official Microsoft Le...
- ☒ **microsoft_docs_fetch** Fetch and convert a Microsoft Learn documentation webpage to markdo...
- ☒ **microsoft_docs_search** Search official Microsoft/Azure documentation to find the most relevan...

GitHub Pull Requests

- ☐ **Mermaid Chat Features**

ghcp-labs workst

ue_fetch, githul

-request-github,

.vscode-pull-rec

ME and source co



But today's
MCP star will
be...

Github MCP Server



Gives AI agents, assistants, and chatbots the ability to read repositories and code files, manage issues and PRs, analyze code, and automate workflows. All through natural language interactions.

Use Cases

- Repository Management: Browse and query code, search files, analyze commits, and understand project structure across any repository you have access to.
- Issue & PR Automation: Create, update, and manage issues and pull requests. Let AI help triage bugs, review code changes, and maintain project boards.
- CI/CD & Workflow Intelligence: Monitor GitHub Actions workflow runs, analyze build failures, manage releases, and get insights into your development pipeline.
- Code Analysis: Examine security findings, review Dependabot alerts, understand code patterns, and get comprehensive insights into your codebase.
- Team Collaboration: Access discussions, manage notifications, analyze team activity, and streamline processes for your team.

GitHub Copilot Spaces



Copilot Spaces is a curated context container — a named, shareable workspace where you assemble the code, docs, issues, PRs, and instructions that ground Copilot's responses in **your specific project context**, rather than generic training data.

What you can add to a Space:

-  Repositories
-  Individual files
-  GitHub Issues
-  Pull Requests
-  Free text / notes
-  Images & uploads

Github Copilot Spaces will allow you to:



- **Unified context across your SDLC**

Brings together repos, files, issues, PRs, and custom instructions into a single, reusable context layer

- **Seamless access from your IDE**

Work directly in VS Code using Agent Mode + MCP — no context switching required

- **Consistent engineering standards at scale**

Applies shared instructions and patterns across teams and repositories

- **Accelerated onboarding**

Helps new developers ramp up quickly without relying on senior engineers

- **Centralised knowledge hub**

Enables cross-team collaboration with shared context and reusable knowledge

- **End-to-end planning from context**

Create and manage issues and projects grounded in real repository context

- **Higher-quality, grounded outputs**

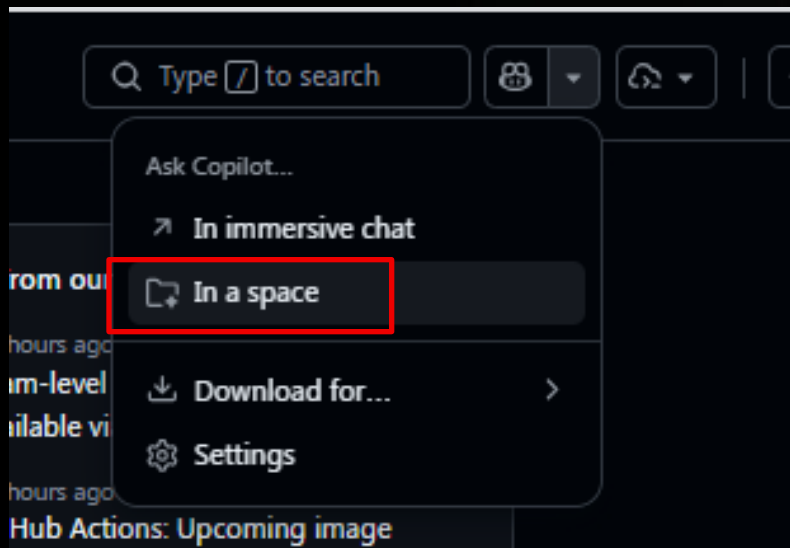
Ensures responses are based on your codebase and workflows — not generic AI outputs

Setting up Spaces for success



Find your space at: <https://github.com/copilot/spaces>

Or on the upper right part of your screen at:

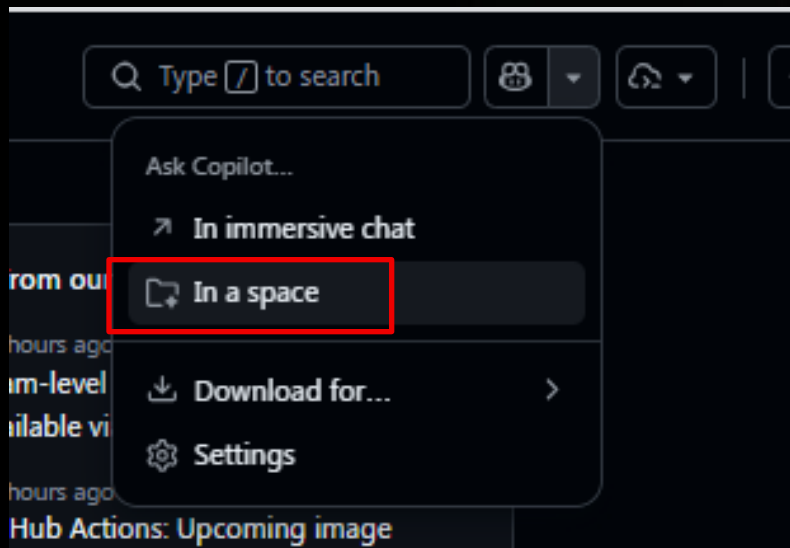


Setting up Spaces for success



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Or on the upper right part of your screen at:





Demo



Now let's connect it to the IDE!



Lab 04

Context Engineering





Let's get back to our repo...

<https://github.com/martaldsantos/ghcp-labs>



Thank you